Deep Learning - COSC2779/2972

Introduction to Deep Learning

A/Prof. Ruwan Tennakoon

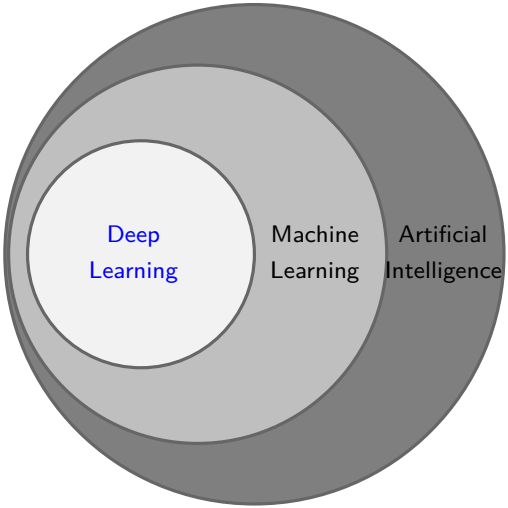
RMIT UNIVERSITY

Reference: *Chapter 5: Ian Goodfellow et. al., "Deep Learning", MIT Press, 2016.*

Outline

1. What is Deep Learning?
2. Deep Learning Applications
3. Machine Learning Basics

What is Deep Learning?



AI: Any technique that enable computers to mimic human behaviour.

ML: Machine learning is the field of study that gives the Ability to learn without being explicitly programmed - Arthur Samuel (1959).

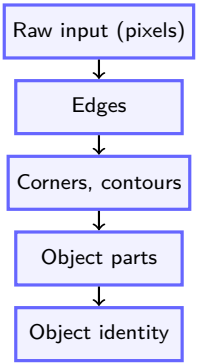
More technically: "A computer program is said to learn:

- Some class of **tasks T**
- From **experience E**, and
- **Performance measure P**

If its performance at tasks in T, as measured by P, improves with experience E".

- Tom Mitchell (1998)

What is Deep Learning?



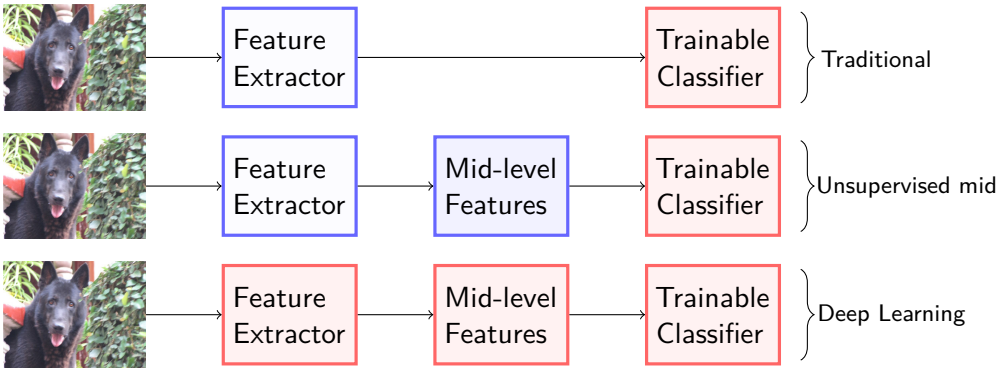
After Goodfellow et al. (2016), Fig. 1.2.

Deep learning is a ML technique that learns features and tasks directly from data using a neural network framework.

"*Deep learning methods aim at* ***learning feature hierarchies*** *with features from higher levels of the hierarchy formed by the composition of lower level features. Automatically learning features at multiple levels of abstraction allow a system to learn complex functions mapping the input to the output* ***directly from data, without depending completely on human-crafted features.***"

- Yoshua Bengio, "Deep learning of representations for unsupervised and transfer learning", 2012.

Learning Hierarchical Representations



Handcrafted features are **time consuming, brittle and not scalable** in practice. DL learn underlying features directly from data.

What is Deep Learning?

According to Yann LeCun, Yoshua Bengio & Geoffrey Hinton, "Deep Learning", Nature 2015.

- Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction.
- Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm to indicate how a machine should change its internal parameters that are used to compute the representation in each layer from the representation in the previous layer.
- Deep convolutional nets have brought about **breakthroughs in processing images, video, speech and audio, whereas recurrent nets have shone light on sequential data such as text and speech.**

Why use Deep Multi Layered Models?

Theoretical result [Cybenko, 1989]: one hidden-layer NN can approximate any continuous function over compact domain to arbitrary accuracy given enough hidden units!.

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Why use Deep Multi Layered Models?

- Argument 1: Visual scenes are hierarchically organized.
- Argument 2: Biological vision is hierarchically organized.
- Argument 3: Shallow representations are inefficient at representing highly varying functions.

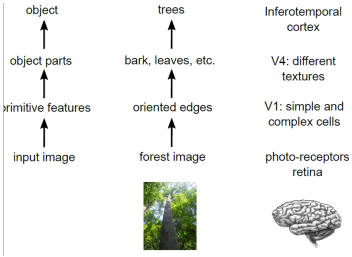
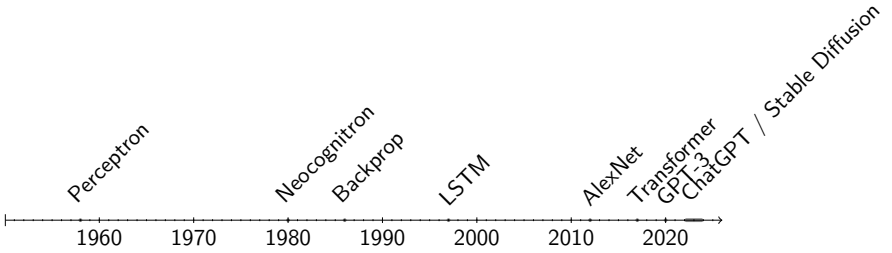


Image: Richard E. Turner

Some History



Neural Networks date back decades, so why the resurgence — and why the second wave (2017–present) of foundation models?

Why Now?

Big Data
Larger datasets.
Easier collection and storage.

IMAGENET

Computation
GPUs and TPUs.
Massively parallelisable.



Software
Improved algorithms.
Open-source frameworks.

TensorFlow
PyTorch

Scale
Scaling laws: loss falls predictably as parameters, data and compute grow (Kaplan 2020, Hoffmann 2022).

Drives the foundation-model era.

Why Now?

The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) is an annual competition helped between 2010 and 2017.

The datasets comprised approximately 1 million images and 1,000 object classes.

The annual challenge focuses on multiple tasks for image classification.

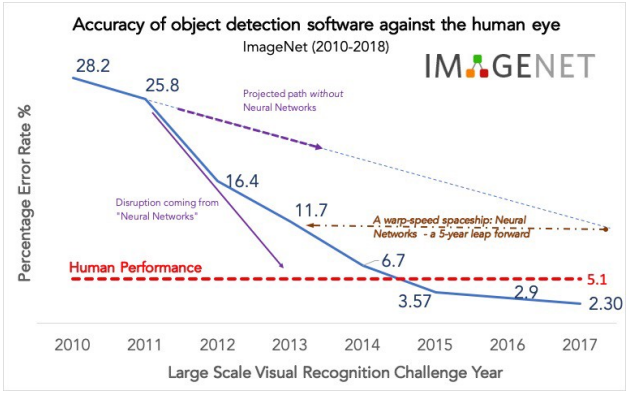
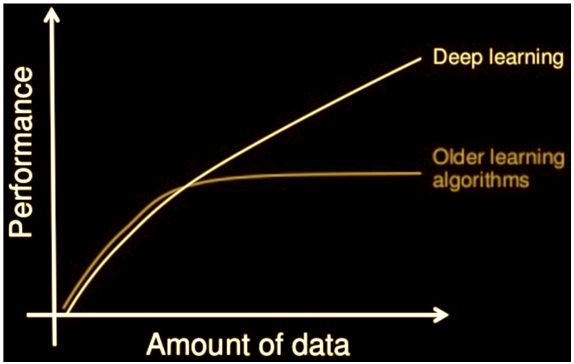


Image source: ImageNet
Alex Krizhevsky, et al. "ImageNet Classification with Deep Convolutional Neural Networks" developed a convolutional neural network that achieved top results on the ILSVRC-2010 and ILSVRC-2012 image classification tasks.

Why Now?

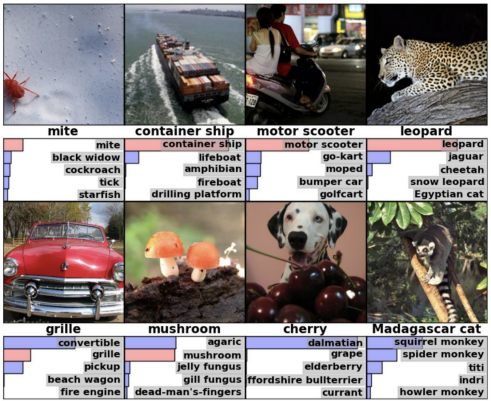


When a large neural networks is trained with more and more data, their performance continues to increase. This is generally different to other machine learning techniques that reach a plateau in performance. - Andrew Ng.

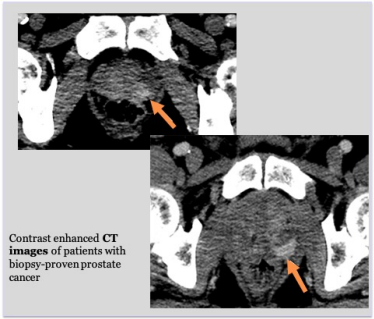
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ImageNet



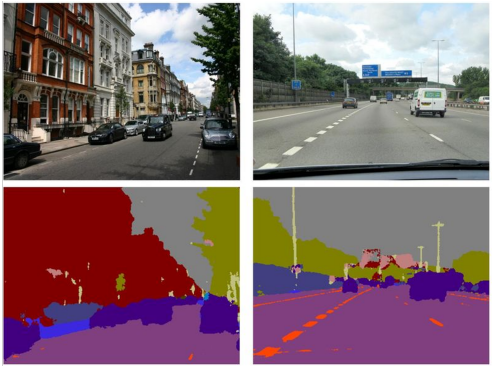
Early Detection of Prostate Cancer



Korevaar, S., Tennakoon, R., Page, M., Brotchie, P., Thangarajah, J., Florescu, C., Sutherland, T., Kam, N.M. and Bab-Hadiashar, A., 2021. Incidental detection of prostate cancer with computed tomography scans. Scientific Reports.

7 News clip

SegNet

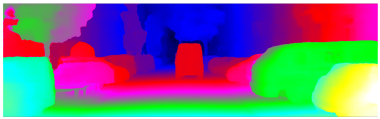


https://mi.eng.cam.ac.uk/projects/segnet/

Depth Estimation - Autonomous Navigation

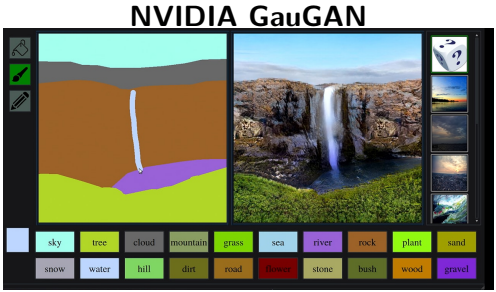


Image



Distance from camera or Depth

Chuah, W., Tennakoon, R., Hoseinnezhad, R. and Bab-Hadiashar, A., 2021. Deep Learning-Based Incorporation of Planar Constraints for Robust Stereo Depth Estimation in Autonomous Vehicle Applications. IEEE Transactions on Intelligent Transportation Systems.

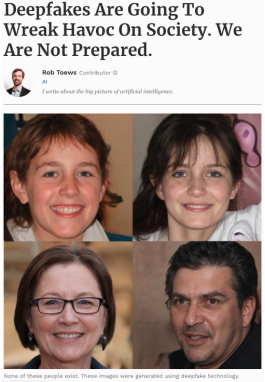


GauGAN: Changing Sketches into Photorealistic Masterpieces

Interactive Demo

Computer Vision - Other

- Object Detection
- Style Transfer
- Image synthesis
- 3D point cloud analysis and scene understanding
- Many more . . .



Forbes article on Deep Fake
It's Getting Harder to Spot a Deep Fake Video

Foundation Models & Generative AI

Large models pre-trained on broad data and then adapted to many downstream tasks — the dominant paradigm since 2020.

- **Large Language Models:** GPT-4 / Claude / Gemini — chat, coding, reasoning, agentic workflows.
- **Image generation:** Stable Diffusion, Midjourney, DALL-E — text-to-image via denoising diffusion.
- **Video and multimodal:** Sora, Veo — text-to-video; vision-language models that read images and documents.
- **Science:** AlphaFold (protein structure), GNoME (materials), weather forecasting (GraphCast).
- **Code and agents:** Copilot, Claude Code — writing and editing software, calling tools, browsing.

A single pre-trained model is reused across many tasks via prompting, fine-tuning, or PEFT (e.g. LoRA).

Capabilities scale predictably with parameters, data and compute — *scaling laws* (Kaplan 2020; Hoffmann 2022).

We will return to transformers, pre-training, and adaptation later in the course.

- Translation
- Speech recognition
- Transcription
- Speech synthesis

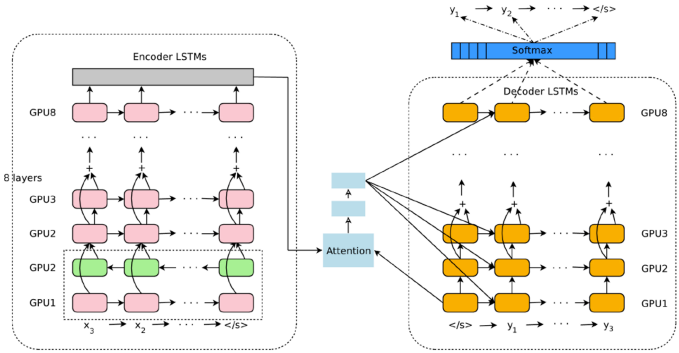


Image: Googles Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

Many Other Fields

- Deep learning for biology
- Applications of machine learning in drug discovery and development
- Deep Learning for Physical Sciences - Radio astronomy, Jet Physics, Quantum physics.
- Weather Forecasting